

RECOVERY TRIAL PROTOCOL

This protocol describes the RECOVERY Trial, a randomised platform trial among patients hospitalised for pneumonia caused by influenza and other pathogens (its full title, Randomised Evaluation of CCOVID-19 ThErapY, reflects its initial focus on COVID-19 alone when it opened in March 2020).

Background: In early 2020, as the RECOVERY trial was being set up, there were no approved treatments for COVID-19, a disease induced by the novel coronavirus SARS-CoV-2 that emerged in China in late 2019. Opening in March 2020, RECOVERY evaluated twenty SARS-CoV-2 therapies, providing reliable evidence about their efficacy and safety that has informed the treatment of patients worldwide. Since then, the progress in COVID-19 treatment has highlighted the need for better evidence for the treatment of pneumonia caused by other pathogens, such as influenza and bacteria, for which therapies are widely used without good evidence of benefit or safety.

Eligibility and randomisation: This protocol (version 28.0) includes treatment comparisons for influenza and community-acquired pneumonia. No COVID-19 comparisons are currently open in the trial. Eligible patients are randomly allocated between one or more treatment arms, each to be given in addition to the usual standard of care in the participating hospital. The study is dynamic, and treatments are added and removed as results and suitable treatments become available, or as new infectious respiratory threats emerge. The randomised treatment comparisons in this version of the protocol (which should be checked and confirmed as the current version) are shown in Table 1. For patients for whom not all the trial arms are appropriate or at locations where not all are available, randomisation will be between fewer comparisons.

Table 1: Current comparisons

Condition	Randomised comparisons, each vs. usual care alone	Eligibility criteria specific to comparison
Influenza	Baloxavir marboxil	No baloxavir marboxil use for current infection
	Oseltamivir	No neuraminidase inhibitor use for current infection
	Corticosteroids	hypoxia; without suspected or confirmed SARS-CoV-2 infection; no corticosteroid use for current infection
Community-acquired pneumonia	Corticosteroids	without suspected or confirmed SARS-CoV-2, influenza, TB ^a or PJP ^b ; no corticosteroid use for current infection
^a active pulmonary tuberculosis; ^b <i>Pneumocystis jirovecii</i> pneumonia (also known as PCP)		

See Appendix 6 for details of the active comparisons in each participating country, and for region-specific information including age, pregnancy and breastfeeding restrictions. Information on completed comparisons is in Section 7.

In a partial factorial design, participants may be entered into one or more randomised comparisons of active treatment plus usual care vs. usual care alone, simultaneously. This allows the effects of one treatment to be assessed in the presence or absence of another which generates useful information for clinicians and health policy-makers. In particular, this allows antiviral therapies to be assessed as monotherapy and in combination, which will provide important information on efficacy, safety and the development of resistance. This protocol indicates clearly where specific combinations are not desirable.

Adaptive design: The interim trial results will be monitored by an independent Data Monitoring Committee (DMC). The most important task for the DMC will be to assess whether the randomised comparisons in the study have provided evidence that is strong enough (with a range of uncertainty around the results that is narrow enough) to affect national and global treatment strategies. In such a circumstance, the DMC will inform the Trial Steering Committee who will make the results available to the public and amend the trial arms accordingly. Regardless, follow-up will continue for all randomised participants, including those previously assigned to trial arms that are modified or ceased. New trial arms can be added as evidence emerges that other candidate therapeutics should be evaluated.

Outcomes: The main outcomes will be death, discharge, need for ventilation and need for renal replacement therapy. For the main analyses, follow-up will be censored at 28 days after randomisation. Additional information on longer term outcomes may be collected through review of medical records or linkage to medical databases where available (such as those managed by NHS England and equivalent local, regional or national organisations).

Simplicity of procedures: To facilitate collaboration, even in hospitals that suddenly become overloaded, patient enrolment (via the internet) and all other trial procedures are greatly streamlined. Informed consent is simple and data entry is minimal. Randomisation via the internet is simple and quick, at the end of which the allocated treatment is displayed on the screen and can be printed or downloaded. Key follow-up information is recorded at a single timepoint and may be ascertained by contacting participants in person, by phone or electronically, or by review of medical records and databases.

Data to be recorded: At randomisation, information will be collected on the identity of the randomising clinician and of the patient, age, sex, major co-morbidity, pregnancy, illness onset date and severity, and any contraindications to the study treatments. The main outcomes will be death (with date and probable cause), discharge (with date), need for ventilation (with number of days recorded) and need for renal replacement therapy. Other information to be recorded relevant to safety will include acute kidney or liver injury, cardiac arrhythmia, infection, thrombosis, bleeding, metabolic disturbances, and seizures.

Reminders will be sent if outcome data have not been recorded by 28 days after randomisation. Suspected Serious Adverse Reactions (SSARs) to one of the study medications (e.g., Stevens-Johnson syndrome, anaphylaxis, aplastic anaemia) will be collected, and unexpected SSARs (SUSARs) will be reported in an expedited fashion.

Other adverse events will not be recorded but may be available through linkage to medical databases.

Numbers to be randomised: The larger the number randomised the more accurate the results will be but the numbers that can be randomised will depend critically on the epidemiology of the relevant infections over the next few years. If substantial numbers are hospitalised in the participating centres then it may be possible to randomise several thousand with mild disease and a few thousand with severe disease, but realistic, appropriate sample sizes could not be estimated at the start of the trial.

Heterogeneity between populations: If sufficient numbers are studied, it may be possible to generate reliable evidence in certain patient groups (e.g. those with major co-morbidity or who are older). To this end, data from this study may be combined with data from other trials of treatments for viral or bacterial pneumonia.

Add-on studies: Particular countries or groups of hospitals, may want to collaborate in adding further measurements or observations, such as serial blood gases or chemistry, serial lung imaging, or serial documentation of other aspects of disease status. While well-organised additional research studies of the natural history of the disease or of the effects of the trial treatments could be valuable (although the lack of placebo control may bias the assessment of subjective side-effects, such as gastrointestinal problems), they are not core requirements.

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www.recoverytrial.net

To enquire about the trial outside of the UK, contact the relevant Clinical Trial Units

To RANDOMISE a patient, visit: www.recoverytrial.net

1 BACKGROUND AND RATIONALE.....	5
1.1 SETTING.....	5
1.2 TREATMENT OPTIONS	5
1.3 MODIFICATIONS TO THE NUMBER OF TREATMENT COMPARISONS	5
1.4 DESIGN CONSIDERATIONS.....	5
1.5 POTENTIAL FOR EFFECTIVE TREATMENTS TO BECOME AVAILABLE	6
2 DESIGN AND PROCEDURES.....	7
2.1 ELIGIBILITY	7
2.2 CONSENT	8
2.3 BASELINE INFORMATION	9
2.4 RANDOMISED ALLOCATION OF TREATMENT FOR INFLUENZA	10
2.5 RANDOMISED ALLOCATION OF TREATMENT FOR COMMUNITY-ACQUIRED PNEUMONIA.....	11
2.6 ADMINISTRATION OF ALLOCATED TREATMENT	11
2.7 COLLECTING FOLLOW-UP INFORMATION	11
2.8 DURATION OF FOLLOW-UP	13
2.9 WITHDRAWAL OF CONSENT	13
3 STATISTICAL ANALYSIS	13
3.1 OUTCOMES	13
3.2 METHODS OF ANALYSIS.....	14
4 DATA AND SAFETY MONITORING	15
4.1 RECORDING SUSPECTED SERIOUS ADVERSE REACTIONS.....	15
4.2 CENTRAL ASSESSMENT AND ONWARD REPORTING OF SUSARs	16
4.3 RECORDING SAFETY INFORMATION AND OTHER ADVERSE EVENTS	16
4.4 ROLE OF THE DATA MONITORING COMMITTEE (DMC)	16
4.5 BLINDING.....	17
5 QUALITY MANAGEMENT	17
5.1 QUALITY BY DESIGN PRINCIPLES.....	17
5.2 TRAINING AND MONITORING	18
5.3 DATA MANAGEMENT	18
5.4 SOURCE DOCUMENTS AND ARCHIVING	18
6 OPERATIONAL AND ADMINISTRATIVE DETAILS.....	19
6.1 SPONSOR AND COORDINATION	19
6.2 FUNDING	19
6.3 INDEMNITY.....	19
6.4 LOCAL CLINICAL CENTRES	19
6.5 SUPPLY OF STUDY TREATMENTS.....	19
6.6 END OF TRIAL	20
6.7 PUBLICATIONS AND REPORTS	20
6.8 SUBSTUDIES.....	20
7 VERSION HISTORY.....	21
8 APPENDICES.....	24
8.1 APPENDIX 1: INFORMATION ABOUT THE TREATMENT ARMS	24
8.2 APPENDIX 2: DRUG SPECIFIC CONTRAINDICATIONS AND CAUTIONS	26
8.3 APPENDIX 3: PAEDIATRIC DOSING INFORMATION.....	27
8.4 APPENDIX 4: USE OF IMPs IN PREGNANT AND BREASTFEEDING WOMEN	28
8.5 APPENDIX 5: ORGANISATIONAL STRUCTURE AND RESPONSIBILITIES.....	29
8.6 APPENDIX 6: ELIGIBILITY BY TRIAL REGION, AGE, AND PREGNANCY/BREASTFEEDING	31
8.7 APPENDIX 7: SCHEDULE OF ASSESSMENTS	31
8.8 APPENDIX 8: ABBREVIATIONS	32
9 REFERENCES	33
10 CONTACT DETAILS.....	35

1 BACKGROUND AND RATIONALE

1.1 Setting

In 2019 a novel coronavirus-disease (COVID-19) emerged in Wuhan, China. A month later the Chinese Center for Disease Control and Prevention identified a new beta-coronavirus (SARS coronavirus 2, or SARS-CoV-2) as the aetiological agent.¹ The clinical manifestations of COVID-19 ranged from asymptomatic infection or mild, transient symptoms to severe viral pneumonia with respiratory failure. Many countries experienced a major increase in the number of hospitalised pneumonia patients, and mortality in hospitalised patients was as high as 30%.²⁻⁴ The RECOVERY platform trial was initially set up to evaluate treatments for this new disease.

Since the RECOVERY trial began in 2020, it has identified several life-saving treatments for COVID-19, and shown that other widely used treatments were ineffective.⁵⁻¹⁴ In contrast, the treatment of hospitalised patients with pneumonia caused by influenza or bacterial infection has progressed little in the last 20 years and there is substantial uncertainty and disagreement about optimal treatment of these patients. Corticosteroids reduce the risk of death in patients with severe COVID-19, but there is insufficient evidence to know if they produce a similar benefit in influenza and bacterial infection.^{15,16} Anti-SARS-CoV-2 antivirals can improve outcomes in hospitalised COVID-19 patients, but there is no similar evidence for anti-influenza antivirals.^{17,18}

1.2 Treatment Options

The protocol allows reliable assessment of the effects of multiple different treatments (including re-purposed and novel drugs) on major outcomes in pneumonia. All patients will receive usual care for the participating hospital. The current treatments under evaluation are summarised in Table 1 above with further details provided in sections 2.4-2.5 and in Appendices 1-4 (sections 8.1-8.4).

1.3 Modifications to the number of treatment comparisons

Other treatment comparisons can be added if evidence emerges that there are suitable candidate therapeutics. Conversely, in some patient populations, not all trial comparisons are appropriate (e.g. due to contraindications based on co-morbid conditions or concomitant medication); in some hospitals or countries, not all treatments will be available (e.g. due to manufacturing and supply issues); and at some times, not all treatments will be active (e.g. due to lack of relevant approvals and contractual agreements). The Trial Steering Committee may elect to pause one or more of the comparisons in order to increase trial efficiency during a fluctuating epidemic. In any of these situations, randomisation will be between fewer arms. Depending on the availability and suitability of treatments, it may be allowed for participants to be randomised in only one or two parts of the main randomisations.

1.4 Design Considerations

The RECOVERY Protocol describes an overarching trial design to provide reliable evidence on the efficacy of candidate therapies for patients hospitalised with pneumonia receiving usual standard of care. Pneumonia is a common cause of hospital admission, particularly during seasonal respiratory virus epidemics, and carries a substantial risk of death, so even treatments with only a moderate impact on survival or on hospital resources could be

worthwhile. Therefore, the focus of RECOVERY is the impact of candidate treatments on mortality and on the need for hospitalisation or ventilation.

Critically, the trial is designed to minimise the burden on front-line hospital staff working within an overstretched care system during a major epidemic. Eligibility criteria are therefore simple and trial processes (including paperwork) are minimised.

The protocol is deliberately flexible so that it is suitable for a wide range of settings, allowing:

- a broad range of patients to be enrolled in large numbers;
- randomisation between only those treatment arms that are *both* available at the hospital *and* not believed by the enrolling doctor to be contraindicated (e.g. by particular co-morbid conditions or concomitant medications);
- treatment arms to be added or removed according to the emerging evidence; and
- additional substudies may be added to provide more detailed information on side effects or sub-categorisation of patient types but these are not the primary objective and are not required for participation.

For influenza and community-acquired pneumonia, the average length of hospital stay is around 7-9 days, and most mortality occurs within 28 days of admission, so assessment at 28 days will capture most outcomes.^{19,20}

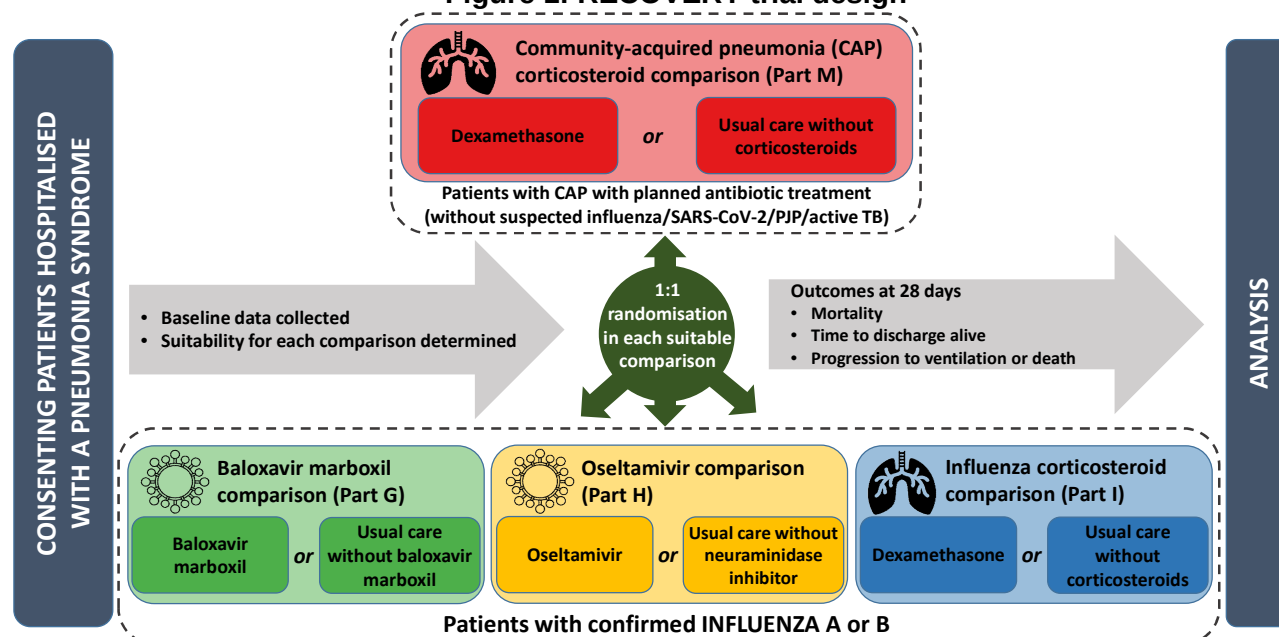
1.5 Potential for effective treatments to become available

In early 2020, when the trial first started, there were no known treatments for COVID-19. However, over time, effective treatments became available, typically as the result of reliable information from randomised trials (including from this study). For example, in June 2020, results from the RECOVERY trial showed that dexamethasone 6mg once daily reduces the mortality in COVID-19 patients requiring mechanical ventilation or oxygen.⁶ In response, many clinical guidelines now recommend the use of dexamethasone 6mg once daily as standard of care for these types of patients.

The RECOVERY trial randomises eligible participants to usual standard of care for the local hospital alone vs usual standard of care plus one or more additional study treatments. Over time, usual standard of care has evolved (e.g. as a consequence of results from trials such as RECOVERY) and it is anticipated that it will evolve further in the future. Thus randomisation will always be relevant to the current clinical situation and the incremental effects of the study treatments will be appropriately assessed.

2 DESIGN AND PROCEDURES

Figure 1: RECOVERY trial design



A summary of trial procedures is provided in Appendix 7 (schedule of assessments).

2.1 Eligibility

Patients are eligible for the study if all of the following are true:

(i) Hospitalised

(ii) Pneumonia syndrome

In general, pneumonia should be suspected when a patient presents with:

- typical symptoms of a new respiratory tract infection (e.g. influenza-like illness with fever and muscle pain, or respiratory illness with cough and shortness of breath); and
- objective evidence of acute lung disease (e.g. consolidation or ground-glass shadowing on X-ray or CT, hypoxia, or compatible clinical examination); and
- alternative causes have been considered unlikely or excluded (e.g. heart failure).

However, the diagnosis remains a clinical one based on the opinion of the managing doctor (the above criteria are just a guide).

(iii) One of the following diagnoses:

- Confirmed influenza A or B infection (including patients with SARS-CoV-2 co-infection)
- Community-acquired pneumonia (CAP) with planned antibiotic treatment (excluding patients with suspected or confirmed SARS-CoV-2, influenza, active pulmonary tuberculosis or *Pneumocystis jirovecii* pneumonia)

(iv) No medical history that might, in the opinion of the attending clinician, put the patient at significant risk if he/she were to participate in the trial

Patients with suspected or confirmed active pulmonary tuberculosis or *Pneumocystis jirovecii* pneumonia (also known as PCP or PJP) are excluded from the CAP comparison, as these infections are caused by specific organisms with distinct pathologies, and so are not usually categorised as CAP. Eligibility for the CAP comparison also requires planned antibiotic treatment, so patients being treated solely for fungal or viral pneumonia are not eligible.

Patients with SARS-CoV-2 and influenza co-infection are eligible, but would be excluded from certain comparisons (as described in the table on page 1). In addition, if the attending clinician believes that there is a specific contra-indication to one of the active drug treatment arms (see Appendix 2, Appendix 3 for children, and Appendix 4 for pregnant and breastfeeding women) or that the patient should definitely be receiving one of the active drug treatment arms then that arm will not be available for randomisation for that patient. For patients who lack capacity, an advanced directive or behaviour that clearly indicates that they would not wish to participate in the trial would be considered sufficient reason to exclude them from the trial.

Patients who have been previously recruited into RECOVERY are eligible to be recruited again as long as their previous randomisation was >6 months ago. Patients will not be recruited into the same randomised comparison (e.g. sotrovimab vs. usual care) on more than one occasion, regardless of how far apart they occur.

In some locations, children (aged <18 years) will not be recruited, to comply with local and national regulatory approvals (see Appendix 6).

2.2 Consent

Informed consent should be obtained from each patient 16 years and over before enrolment into the study. If a patient does not have capacity to consent due to severe disease (e.g. needing ventilation) or a prior condition, then legal representative consent is required for the patient to participate in the trial. Due to the poor outcomes in patients hospitalised with pneumonia (5-10% mortality, rising to over 40% in those with severe disease^{21,22}), patients who lack capacity to consent and for whom a relative to act as the legally designated representative is not available (in person), randomisation and consequent treatment can proceed with consent provided by a clinician (independent of the trial^a) who can act as the legally designated representative (if allowed by local regulations).

If they regain capacity, such participants should be provided with information about the trial (ideally prior to discharge, but otherwise as soon as possible thereafter), what their rights are and how to exercise them, but it is not necessary to obtain their written consent^b. Provision of such information (i.e. the current participant information sheet) will be documented in the medical record.

For children aged <16 years old consent will be sought from their parents or legal guardian. Where possible, children aged between 10-15 years old will also be asked for assent.

^a Independent clinicians may complete study training, but have no other involvement in the trial, e.g. eligibility assessment, or randomisation

^b Unless required by local regulations. (This is not required in the UK.)

Children aged ≥ 16 years old will be asked for consent as for adults. Witnessed^c consent may be obtained over the telephone or web video link if hospital visiting rules or parental infection mean a parent/guardian cannot be physically present.

Information about participants' involvement will be included in routine clinical communications (e.g. discharge summaries) provided to participants (and, in the UK their GPs). If any other relevant information arises during the trial, this may also be sent to GPs.

2.3 Baseline information

The following information will be recorded on the web-based form by the attending clinician or delegate:

- Patient details (depending on local privacy requirements, this may include name or initials, NHS/CHI number [UK only] or medical records number, date of birth, sex)
- Clinician details (e.g. name)
- Symptom onset date
- Disease severity as assessed by need for supplemental oxygen, non-invasive ventilation or invasive mechanical ventilation/extracorporeal membrane oxygenation (ECMO)
- Oxygen saturations on air, oxygen delivery device, and oxygen flow rate (if available)
- Latest routine measurement of respiratory rate and blood pressure
- Presence of new or worsened confusion
- Presence of lung consolidation on chest X-ray or CT (if available)
- Latest routine measurement of creatinine, urea, C-reactive protein, and procalcitonin (if available)
- Influenza test results (if available, and note confirmation of influenza infection is required for entry into influenza comparisons)
- Major co-morbidity (e.g. heart disease, diabetes, chronic lung disease) and pregnancy (including pregnancy test result in all women of child-bearing potential^d)
- Use of relevant medications (e.g. corticosteroids, antivirals) and prior vaccination
- Date of hospitalisation
- Contraindication to the study treatment regimens (in the opinion of the attending clinician)
- Name of person completing the form

The person completing the form will then be asked to confirm that they wish to randomise the patient and will then be required to enter their name and e-mail address.

^c The witness should be impartial i.e. not a member of the research team, but they do not require specific training or knowledge of the trial.

^d A woman of childbearing potential is defined as a post-menarchal pre-menopausal female capable of becoming pregnant. This includes women on oral, injectable, or mechanical contraception; women who are single; women whose male partners have been vasectomized or whose male partners have received or are utilizing mechanical contraceptive devices. The potential inclusion of any pregnant women should be discussed with a consultant obstetrician (or obstetric physician).

2.3.1 Baseline sample collection^e

2.3.1.1 Participants with influenza pneumonia

Participants in the UK with influenza pneumonia should have a nasal swab collected in which the presence of influenza virus will be measured. Participants outside the UK do not require baseline sample collection, although if the influenza diagnosis was based on a rapid antigen test alone then a nose or throat swab will be collected for influenza PCR at a clinical laboratory (not at sites in the EU, and only if this testing is locally available). This swab will be collected after obtaining consent and prior to randomisation, and patients with a positive antigen test may proceed to randomisation before results of influenza PCR are available.

2.3.1.2 Participants with community-acquired pneumonia

These participants do not require any samples to be collected.

2.4 Randomised allocation of treatment for influenza

In addition to receiving usual care, eligible patients with confirmed influenza A or B infection will be allocated treatment(s) using a central web-based randomisation service (without stratification or minimisation). A factorial design is used such that eligible patients may be randomised simultaneously to one or more of the study treatment arms (depending on location and infection). The doses in this section are for adults (see Appendix 3 for paediatric dosing). Region-specific exclusions, including those related to age, pregnancy or breastfeeding, are given in Appendix 6.

2.4.1 Randomisation part G

Eligible patients (with or without SARS-CoV-2 co-infection) may be randomised in a ratio of 1:1 to one of the arms listed below.

- **No additional treatment**
- **Baloxavir marboxil 40mg (or 80mg if weight ≥ 80 kg) once daily** by mouth or nasogastric tube to be given on day 1 and day 4^f.

2.4.2 Randomisation part H

Eligible patients (with or without SARS-CoV-2 co-infection) may be randomised in a ratio of 1:1 to one of the arms listed below.

- **No additional treatment**
- **Oseltamivir 75mg twice daily** by mouth or nasogastric tube for five days^{f,g}.

^e Collection of these samples will continue until the Steering Committee determine (on the basis of data blinded to treatment allocation) that sufficient information is available to assess the effect of treatment on viral load and/or resistance markers.

^f If participant is discharged before course is complete, the participant should be provided with medication to complete the course at home.

^g Course can be extended to 10 days for immunosuppressed patients at the managing clinician's discretion.